ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

Course Code	CCE2233	**Course Name**	Requirements Engineering
Lecturer	Mdm. Nadiah Arsat, Ts. Mohd Zulkifli bin Mohd Zaki		
Semester	2, 2025/2026		
Assessment	Group Project		
Weightage (%)	20		
Marks	100		
Course Learning Outcome	CLO3: Practice ethically in handling and managing requirements artefacts (A2, PLO11)		

Group No.	12
Project Topic	AIU Student Hub
Group Members Name & student ID	1. Eilaf Salaheldeen Ahmed Mohamed - AIU24102053 2. Chit Ko Ko - AIU24102180 3. HASNAIN HAIDER - AIU24102102 4. S M NADIM MAHMUD - AIU24102398
Total Marks (100)	

Project Rubrics

No.	Criteria	Weight	Exemplary (4 points)	Proficient (3 points)	Developing (2 points)	Beginning (1 point)	
Report (50 Marks)							
1	Introduction (Scope and Objectives)	6 Marks	Clear project context, relevant real-world system, and well-defined scope	Clear topic and objectives with minor gaps	Basic description, unclear scope/objectives	Poor or missing introduction	
2	Methodology	10 Marks	Appropriate methods clearly explained; stakeholders well-identified; ethical data collection clearly described	Methods and stakeholders mostly clear; some ethical mention	Limited methods or unclear stakeholders; weak ethical explanation	Missing or incorrect methodology	
3	Requiremen t Specification	10 Marks	Complete and well-structured FR & NFR; clear, consistent format; appropriate prioritization technique with strong justification	Mostly complete FR & NFR; prioritization present but basic	Some requirements missing/unclear; weak prioritization	Very limited or missing requirements	
4	Requirement Management	8 Marks	Complete RTM ; clear change tracking and version control evidence	Mostly complete RTM; basic change tracking	Partial RTM; unclear change management	Missing RTM and change tracking	
5	Ethical Practices & Reflection	8 Marks	Clearly identifies ethical issues; explains how handled; demonstrates fairness, transparency, privacy; insightful reflection	Ethical discussion present with some reflection	Limited ethical explanation or reflection	Very weak or missing	

6	References (APA Style)	4 Marks	≥5 credible sources; correct APA format; well-cited throughout	≥5 sources with minor APA errors	<5 sources or inconsistent format	Missing or incorrect referencing	
7	Organization & Structure	4 Marks	Excellent structure, clear headings, logical flow, professional writing	Well-organized with minor issues	Some disorganization or unclear flow	Poor structure, difficult to follow	
Presentation (40 Marks)							
1	Content & understanding	15 Marks	Covers all key components; demonstrates strong understanding with accurate and well-explained content	Covers most components; explanations generally accurate with minor gaps	Some key components missing; explanations limited or partially incorrect	Very limited content; poor understanding ; major inaccuracies	
2	Organization & Clarity	10 Marks	Well-structured presentation; logical flow; clear transitions; ideas easy to follow	Generally well-organized; minor issues with flow or clarity	Some disorganization; ideas not always clearly connected	Poor structure; difficult to follow; unclear progression	
3	Communication Skills	10 Marks	Confident delivery; clear voice; good pace; effective eye contact; engages audience	Generally clear delivery; minor issues with confidence or clarity	Noticeable hesitation; uneven delivery; limited engagement	Unclear, very hesitant, or difficult to understand	
4	Teamwork & Participation	5 Marks	All members contribute equally; smooth coordination; clear role distribution	Most members participate; coordination generally good	Uneven participation; weak coordination	One or few members dominate; lack of teamwork	
Adab (10 Marks)							
1	Respectful Behaviour	4 Marks	Always respects others' opinions, space, and rights, even in disagreement.	Usually respectful, even under pressure; minor lapses.	Shows some respect but may interrupt or dismiss others.	Rarely demonstrates courtesy or patience.	
2	Cultural Sensitivity	3 Marks	Sensitive and inclusive toward diverse cultural, religious, and social norms.	Acknowledges diversity and avoids bias.	Occasionally missteps in cultural interactions.	Shows discomfort or lack of awareness in diverse settings.	
3	Proper Conduct	3 Marks	Consistently follows etiquette, institutional norms, and moral decorum.	Generally well-behaved and respectful of formal settings.	Needs reminders on proper classroom or workplace behavior.	Often informal or inappropriate in formal settings.	

Table of Contents

Executive Summary

The AIU Student Hub is a centralized academic and logistical platform serving students, lecturers, and management at Albukhary International University (AIU). It functions as a daily attendance system, a communication channel between management departments and students for matters such as results, examinations, course registration, and exemptions, and a medium for student participation in SRC elections. Despite its importance, the existing system suffers from fragmented services, tedious processes, and exploitable security weaknesses. This report documents the requirements for an improved Student Hub based on a three-week requirement elicitation process conducted on the AIU campus. The project set out to apply structured requirements engineering practices to the AIU Student Hub. Specifically, it aimed to gather system requirements using a range of elicitation techniques, document those requirements systematically using established documentation methods, and manage and track changes to the requirements throughout the project lifecycle. To capture a complete picture of stakeholder needs, the team applied data triangulation across three elicitation techniques: an online student survey, semi-structured interviews with student representatives, lecturers, and management staff, and observation-based analysis of the existing Student Hub. The elicitation process yielded 25 functional requirements and 5 non-functional requirements for the AIU Student Hub. The methodological triangulation strategy proved more reliable than any single technique and, alongside careful ethical practice, was central to the project's success.

1. Introduction

1.1 Project background and context

Student Hub is an AIU-based system designed to provide a centralized platform for students, lecturers, and management to perform academic and logistical processes with different departments at AIU. On a day-to-day basis, the hub serves as an attendance platform for students and lecturers. Moreover, the platform also serves as a communication hub for different management departments with students for matters related to results, examinations, course registration, and exemption. The hub also enables students to participate in decision-making by voting in SRC elections. All this makes the student hub an essential, frequently used system in AIU. However, several challenges face stakeholders when dealing with the system, including fragmented services, tedious processes, and exploitable security safeguards. In this report we outline the requirements for the Student Hub based on a 3-week elicitation process on the AIU campus.

1.2 Objectives

- Use different requirements elicitation tools to gather the requirements of the AIU student hub system
- Use proper requirement documentation techniques to systematically document the requirements of the system.
- Use requirement management tools to track the changes in the requirements

2. Methodology

Modern requirements engineering practices emphasize continuous stakeholder involvement and iterative refinement to accommodate evolving organizational and user needs (Zaveri et al., 2024), which guided the structured approach adopted below.

2.1 Elicitation techniques used

The table below maps each technique with the information we hoped to gain from it:

Req. ID	**Elicitation techniques**
Satisfier	Online student survey and semi-structured interviews with student representatives, lecturers, and management staff.urvey
Dissatisfier	Observation-based methods of the existing AIU student hub system.
Delighter	Creative brainstorming exercise in interviews with 2 lecturers and 3 management staff

Table 1. Elicitation techniques

Effective requirements elicitation requires systematic stakeholder engagement and structured information-gathering processes to ensure complete and accurate requirement capture (Hidalgo et al., 2025).

2.2 List of identified stakeholders and descriptions

Stakeholder identification is a critical activity in requirements engineering because different stakeholder groups possess different perspectives, priorities, and expectations regarding system functionality and quality attributes (Elneel et al., 2022).

2.2.1 Students

There are different tracks for students at AIU, including pre-college, undergraduate, or master's. The precollege students are enrolled in either the language center or the school for foundation studies. The undergraduate and master's programs are offered by the School of Business and Social Sciences (SBSS), School of Computing and Informatics (SCI), and School of Education and Human Sciences (SEHS). The AIU student body involves students from more than 60 countries with different digital fluency levels. Students have different expectations around the student hub system. The general expectation is that the system support different dimensions of their student life, including academic and logistical aspects as well as their overall welfare in the university Student survey results and student representative interviews were used to capture common issues such as repeated login, outdated interface design, attendance access, timetable notifications, SRC election security, and privacy concerns.

2.2.2 Lecturers

The lecturers supported by the system include both full-time and part-time lecturers. Each lecturer instructs a different number of courses; some courses are instructed by two lecturers together. The lecturer's needs include having a centralized system to share an attendance code with students. The system shall enable the lecturer to track student attendance.

2.2.3 AIU Management

Management staff at AIU include both high-level managers and staff in managerial positions. These stakeholders are interested in tracking the academic records of all students to have insights about their academic performance.

2.3 Ethical Considerations in Stakeholder Engagement

All elicitation activities followed accepted ethical guidelines for research involving human subjects. Before each interview, participants signed a written informed consent form explaining the study’s purpose, their role, and how their data would be handled; they could withdraw at any time without consequence, and no data from a withdrawn participant was used in the analysis. Participants could choose to remain anonymous or use their real name, and interviews were recorded only with explicit prior agreement, with all data handled to prevent re-identification. The online survey collected no personally identifying information such as names, emails, or student IDs, since anonymity was judged necessary for students to give honest feedback, including criticism of current procedures. A separate ethical issue arose when several survey responses showed signs of AI-generated content—generic, lacking personal context, and not representative of real student experience. Because such responses risked skewing requirement priorities, we addressed this by cross-referencing survey data with observation and interview findings to filter out implausible patterns and ground the analysis in verified input.

3. Requirement Specification

3.1 Functional and non-functional requirements

3.1.1 Functional Requirements

- FR 01: The system shall allow students to register to Student Hub using their student email and ID details.
- FR 02: The system shall allow students to log in to the system using the student username and password.
- FR 03: The system shall allow students to record their attendance for courses they have registered for by scanning the QR code.
- FR 04: The system shall allow students to view their attendance statistics for each course.
- FR 05: The system shall allow students to view their transcript for each semester.
- FR 06: The system shall allow students to register for offered courses.
- FR 07: The system shall allow students to vote on SRC elections using a two-step verification process.
- FR 08: The system shall allow students to request course exemption.
- FR 09: The system shall allow students to renew visas.
- FR 10: The system shall provide a dashboard for management to view student attendance details
- FR 11: The system shall allow management to generate and download student attendance reports by semester or faculty.
- FR 12: The system shall use role-based access control so that each management level can manage student data according to their role.
- FR 13: The system shall send alerts to management when a student's academic performance drops below a set level.
- FR 14: The system shall include an internship management module to record internship details.

- FR 15: The system shall allow lecturers to log in using their staff email and credentials
- FR 16: The system shall provide lecturers with a unified student attendance profile.
- FR 17: The system shall allow lecturers to generate student academic reports automatically.
- FR 18: The system shall allow the lecturer to update student attendance records by filling out a form with student name, course, and date.
- FR 19: The system shall recommend suitable courses based on academic history and previously failed courses.
- FR 20: The system shall notify students when prerequisite courses have not been completed.
- FR 21: The system shall allow lecturers to view advising records for students under their supervision.
- FR 22: The system shall automatically detect timetable conflicts for students during course registration.
- FR 23: The system shall allow students to access services offered by the AIU COF library.
- FR 24: The system shall allow students to view the remaining balance in AIU cafeteria coupon balance
- FR 25: The system shall allow students to view their timetable from the student hub

3.1.2 Nonfunctional Requirements

- NFR 01: The system shall be 99% available on a 24/7 basis.
- NFR 02: The system shall be able to serve >5000 students at the same time.
- NFR 03: The system shall work on both mobile and desktop.
- NFR 04: The system shall handle student data and passwords securely.
- NFR 05: The system shall provide 3 languages of support (Arabic, English, and Malay) for accessibility.

3.2 Requirement Prioritization

Below we map the requirements with their respective priorities using the MoSCoW technique, which is widely used to support effective requirement prioritization and decision-making in software projects involving multiple stakeholders (Vijayakumar et al., 2024). Note that ‘M’ refers to 'must have,' ‘S’ refers to 'should have,' and ‘C’ refers to 'could have.'

Req. ID	Description	MoSCoW Priority	Justification
FR 01	The system shall allow students to register to Student Hub using their student email and ID details.	M	The system cannot work without student registration functionality.
FR 02	The system shall allow students to log in to the system using the student username and password.	M	The system cannot work without student login functionality, as unauthorized access can lead to data manipulation.
FR 03	The system shall allow students to record their attendance for courses they have registered for by scanning the QR code.	M	The attendance recording functionality is the core of the system, so it must be there.
FR 04	The system shall allow students to view their attendance statistics for each course.	S	It's important, but the system can still work without it as attendance will already be recorded.
FR 05	The system shall allow students to view their transcript for each semester.	M	Viewing transcript functionality is the core of the system, so it must be there.
FR 06	The system shall allow students to register for offered courses.	M	Course registration functionality is the core of the system, so it must be there.
FR 07	The system shall allow students to vote on SRC elections using a two-step verification process.	S	It's important, but the system can still work without it. Previously, students would vote manually.
FR 08	The system shall allow students to request course exemption.	S	It's important, but the system can still work without it, as not all students need it.
FR 09	The system shall allow students to renew visas.	S	It's important, but the system can still work without it, as they can use the ISSU form.
FR 10	The system shall provide a dashboard for management to view student attendance details	M	The system cannot work without this dashboard functionality.
FR 11	The system shall allow management to generate and download student attendance reports by semester or faculty.	M	The system cannot work without this reporting functionality.
FR 12	The system shall use role-based access control so that each management level can manage student data according to their role.	M	The system cannot work without this role-based access functionality, as unauthorized access can lead to data leakage.
FR 13	The system shall send alerts to management when a student's	S	It's important to develop this notification functionality, but the system can still work

	academic performance drops below a set level.		without it.
FR 14	The system shall include an internship management module to record internship details.	S	It's important to develop this internship tracking functionality, but the system can still work without it.
FR 15	The system shall allow lecturers to log in using their staff email and credentials	M	The system cannot work without this functionality, as unauthorized access can jeopardize the integrity of academic records.
FR 16	The system shall provide lecturers with a unified student attendance profile.	M	The system cannot work if lecturers cannot assess students' attendance level, as there are consequences for students with attendance below 80%.
FR 17	The system shall allow lecturers to generate student academic reports automatically.	M	The system cannot work without this report generation functionality.
FR 18	The system shall allow the lecturer to update student attendance records by filling out a form with student name, course, and date.	M	The system cannot work without this functionality because sometimes students' attendance needs to be updated from the back office.
FR 19	The system shall recommend suitable courses based on academic history and previously failed courses.	C	This is a nice-to-have feature if time allows.
FR 20	The system shall notify students when prerequisite courses have not been completed.	M	This is an important feature for verifying academic discipline, and the system cannot work without it.
FR 21	The system shall allow lecturers to view advising records for students under their supervision.	S	This is an important feature, but the system can still work without it, as advisees can share their records with the advisor.
FR 22	The system shall automatically detect timetable conflicts for students during course registration.	S	This is an important feature to ensure consistent class assignment, but the system can still work without it.
FR 23	The system shall allow students to access services offered by the AIU COF library.	C	This is a nice-to-have feature if time allows
FR 24	The system shall allow students to view the remaining balance in AIU cafeteria coupon balance	S	This is an important feature, but the system can still work without it, as students can do that at the cafeteria.
FR 25	The system shall allow students to view their timetable from the student hub	S	This is an important feature but the system can still work without it
NFR 01	The system shall be 99% available on a 24/7 basis.	M	Availability is a core quality requirement of the system, and the system cannot work without it.
NFR 02	The system shall be able to serve >5000 students at the same time.	M	The system cannot work if it fails to accommodate all the student body at AIU.

NFR 03	The system shall work on both mobile and desktop.	M	The system cannot work without it because some of the students do not have laptops
NFR 04	The system shall handle student data and passwords securely.	M	The system cannot work without this quality because there are incentives for students to hack the account of others.
NFR 05	The system shall provide 3 languages of support (Arabic, English, and Malay) for accessibility.	C	It's a nice-to-have feature if time allows

4. Requirement Management

4.1 Requirements Traceability Matrix (RTM)

Requirement traceability helps maintain clear relationships between stakeholder needs, documented requirements, and future system changes throughout the software development lifecycle (Mucha et al., 2024).

Req. ID	Description	Source	Related Requirement	Status
FR 01	The system shall allow students to register to Student Hub using their student email and ID details.	Student	N/A	Approved
FR 02	The system shall allow students to log in to the system using the student username and password.	Student	FR 01	Approved
FR 03	The system shall allow students to record their attendance for courses they have registered for by scanning the QR code.	Student	FR 02	Approved
FR 04	The system shall allow students to view their attendance statistics for each course.	Student	FR 03	Approved
FR 05	The system shall allow students to view their transcript for each semester.	Student	FR 02, FR 06	Approved
FR 06	The system shall allow students to register for offered courses.	Student	FR 02	Approved
FR 07	The system shall allow students to vote on SRC elections using a two-step verification process.	Student	FR 02	Proposed
FR 08	The system shall allow students to request course exemption.	Student	FR 02	Rejected
FR 09	The system shall allow students to renew visas.	Student	FR 01	Proposed
FR 10	The system shall provide a dashboard for management to view student attendance details	Management	FR 03	Approved
FR 11	The system shall allow management to generate and download student attendance reports by semester or faculty.	Management	FR 10	Approved
FR 12	The system shall use role-based access control so that each management level can manage student data according to their role.	Management	FR 11	Approved
FR 13	The system shall send alerts to management when a student's academic performance drops below a set level.	Management	FR 05	Proposed

FR 14	The system shall include an internship management module to record internship details.	Managemen t	FR 01	Proposed
FR 15	The system shall allow lecturers to log in using their staff email and credentials	Lecturers	N/A	Approved
FR 16	The system shall provide lecturers with a unified student attendance profile.	Lecturers	FR 15	Approved
FR 17	The system shall allow lecturers to generate student academic reports automatically.	Lecturers	FR 15	Proposed
FR 18	The system shall allow the lecturer to update student attendance records by filling out a form with student name, course, and date.	Lecturers	FR 15	Proposed
FR 19	The system shall recommend suitable courses based on academic history and previously failed courses.	Students	FR 06	Proposed
FR 20	The system shall notify students when prerequisite courses have not been completed.	Students	FR 06	Approved
FR 21	The system shall allow lecturers to view advising records for students under their supervision.	Lecturers	FR 05	Proposed
FR 22	The system shall automatically detect timetable conflicts for students during course registration.	Students	FR 06	Proposed
FR 23	The system shall allow students to access services offered by the AIU COF library.	Students	FR 02	Proposed
FR 24	The system shall allow students to view the remaining balance in AIU cafeteria coupon balance	Students	FR 02	Proposed
FR 25	The system shall allow students to view their timetable from the student hub	Students	FR 02FR 02, FR 06FR 06	Proposed

4.2 Change Management

Version	**Date**	**Description**	**Author**
StudentHub_SRS_v01	May 23, 2026	Created the document structure	Eilaf Salaheldeen Ahmed

			Mohamed
StudentHub_SRS_v02	June 9, 2026	Added Student requirements and placed formatting comments	Eilaf Salaheldeen Ahmed Mohamed
StudentHub_SRS_v03	June 11, 2026	Added the requirement traceability matrix and requirement prioritization.	Eilaf Salaheldeen Ahmed Mohamed
StudentHub_SRS_v04	June 14, 2026	Added management requirements and reported ethical considerations	HASNAIN HAIDER
StudentHub_SRS_v05	June 15, 2026	Edited requirement management and edited traceability matrix for management requirements	HASNAIN HAIDER
StudentHub_SRS_v06	June 15, 2026	Added lecturer requirements and added new student requirements	Chit Ko Ko
StudentHub_SRS_v07	June 16, 2026	Added management requirements in the requirement prioritization matrix	HASNAIN HAIDER
StudentHub_SRS_v08	June 17, 2026	Updated lecturer requirements and added them to the traceability matrix	Chit Ko Ko
StudentHub_SRS_v09	June 18, 2026	Edited justifications in requirement prioritization	S M NADIM MAHMUD
StudentHub_SRS_v10	June 19, 2026	Aligned requirement formatting across different categories (Student, Lecturer, and Management)	S M NADIM MAHMUD

5. Ethical Practices

One ethical issue that emerged during the survey process was that some students wanted to remove the 30-second reset for QR codes. Although this request reflected students' need for a more convenient attendance process, it conflicted with lecturers' need to protect the integrity of attendance records. Balancing convenience against control is a recurring tension in system access design, and the importance of role-based access and controlled information access is widely recognized as a fundamental mechanism for protecting sensitive organizational and user data (Wang et al., 2022).

Another issue was that some survey responses appeared to be AI-generated. This created a serious concern about data validity because those responses were generic and lacked grounded stakeholder experience useful for requirement generation.

Other ethical practices included informed consent for interviews. Each interviewee had read and signed the consent form, and the form asked whether they preferred to participate using their names or as anonymous participants. For anonymous participation, we used secure data handling to avoid revealing participants' identities.

6. Reflection

The main thing we have learned is that the success of requirement engineering depends on stakeholders' engagement. As we struggled to get students to fill out the surveys to help us understand their needs, we realized the elicitation process does not stop at identifying the stakeholders but rather goes a step further to make the engagement attractive for them. And making engagement attractive while being ethical was the hardest challenge we have faced.

The way we tackled it is through data triangulation; instead of relying on one elicitation technique, we combined three tools to maximize the data we can get about the stakeholders. This has helped us to learn resourcefulness and to utilize what we have to do the most good. We also learned a lot about the degree of interpersonal sensitivity needed to manage actual stakeholders. We had to learn how to ask questions that produced requirements without pressuring interviewees to give specific responses because interviewees' experience with requirements engineering varied. More broadly, each team member improved their ability in structured requirement documentation, elicitation, and management.

7. Conclusion

This report presents a structured requirements specification for the AIU Student Hub system, developed through an elicitation process involving students, lecturers, and management personnel. The team identified 25 functional requirements and 5 non-functional requirements covering the system's essential features, such as attendance management, course registration, academic reporting, and security, using a mix of online questionnaires, observational techniques, and semi-structured interviews.

This project's ethical component turned out to be more significant than expected. The rise of AI-generated survey responses highlighted a more general issue with digital participatory research: making sure that the data gathered accurately captures stakeholder experience. In addition to reducing this danger, our approach—methodological triangulation—produced a more comprehensive and convincing list of requirements than any one elicitation method would have. These findings also align with broader digital transformation initiatives occurring in higher education institutions worldwide, where universities increasingly seek to centralize academic and administrative services through integrated digital platforms (Farias-Gaytan et al., 2023).

8. References

Elneel, D. A., Fakharudin, A. S., Ahmed, E. M., Kahtan, H., & Abdullateef, M. (2022). Stakeholder identification overview and challenges in requirements engineering prospective. In Proceedings of the 2022 2nd International Conference on Computing and Information Technology (ICCIT 2022) (pp. 314–319). IEEE. https://doi.org/10.1109/ICCIT52419.2022.9711653

Farias-Gaytan, S., Aguaded, I., Wilberforce, T., & Ramírez-Montoya, M. (2023). Digital transformation initiatives in higher education institutions: A multivocal literature review. Education and Information Technologies, 28, 12351–12382. https://doi.org/10.1007/s10639-022-11544-0

Hidalgo, M., Yanine, F., Paredes, R., Frez, J., & Solar, M. (2025). What is the process? A metamodel of the requirements elicitation process derived from a systematic literature review. Processes, 13(1), 20. https://doi.org/10.3390/pr13010020

Mucha, J., Kaufmann, A., & Riehle, D. (2024). A systematic literature review of pre-requirements specification traceability. Requirements Engineering, 29, 119–141. https://doi.org/10.1007/s00766-023-00412-z

Vijayakumar, S., Krishna Prasad, K., & Raviraja Holla, M. (2024). Assessing the effectiveness of MoSCoW prioritization in software development: A holistic analysis across methodologies. EAI Endorsed Transactions on Internet of Things, 10. https://doi.org/10.4108/eetiot.6515

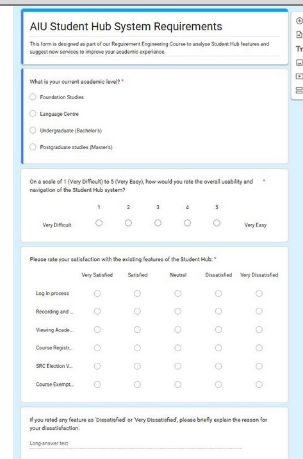
Wang, Y., Li, X., & Chen, Z. (2022). Role-based access control model for inter-system cross-domain in multi-domain environment. Applied Sciences, 12(24), 13036. https://doi.org/10.3390/app122413036

Zaveri, A. A., Jaafar, J., Yafi, E., & Shehmir, S. (2024). Evolution of requirements engineering in agile methodology – Literature review. Journal of Engineering Technology and Applied Physics, 6(2), 57–65. https://doi.org/10.33093/jetap.2024.6.2.9

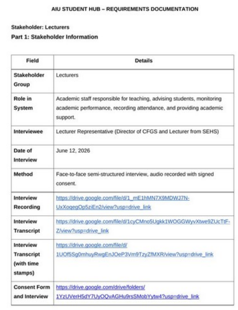
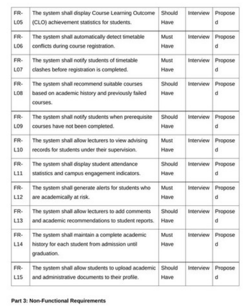
Appendices

The appendices are organized by group member responsibility and include supporting links and visual evidence used during requirement elicitation and documentation.

Appendix 1: Eilaf Salaheldeen Ahmed Mohamed - Student Survey Resources

Resource / Evidence	Link or Location	Visual Evidence
Student survey questionnaire	Open survey form	
Survey response evidence	[Placeholder - insert Form Response Excel Sheet link manually]	
Student survey resource folder	Open resource folder	 Albukhary International University (AIU) CCE2233 – Requirements Engineering INFORMED CONSENT FORM AIU Student Hub – Requirements Engineering Project 1. Purpose You are invited to take part in a group project for the CCE2233 Requirements Engineering course at AIU. The aim is to collect and document requirements for a proposed AIU Student Hub system. Your responses will be used only for academic purposes. 2. What You Will Be Asked to Do You will take part in a short interview (about 15 to 30 minutes). You will be asked about your role, your needs from the system, and your expectations for data privacy and security. 3. Voluntary Participation Your participation is completely voluntary. You may: - Refuse to answer any question - Stop the interview at any time - Request that your responses not be included in the report 4. Confidentiality All information you provide will be kept confidential. Your name will not appear in the project report. You will be referred to only as 'AIU Management Representative'. Data collected will only be accessible to the project group and will be deleted after submission on 18 June 2026. 5. Recording Please indicate whether you agree to be audio recorded during the interview: ☑ I agree to be audio recorded ☐ I do not agree to be audio recorded

Appendix 2: Chit Ko Ko - Lecturer Interview Resources

Resource / Evidence	Link or Location	Visual Evidence
Lecturer resource folder	Open resource folder	
Interview recording	Open recording	
Interview transcript and timestamped transcript	Open transcript folder	
Consent form and interview form	Open consent/interview forms	

AIU STUDENT HUB – REQUIREMENTS DOCUMENTATION

Stakeholder: Lecturers

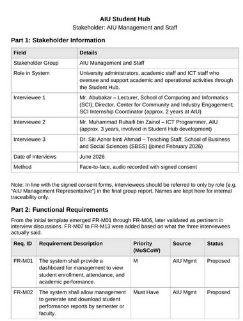
Part 1: Stakeholder Information

Field	Details
Stakeholder Group	Lecturers
Role in System	Academic staff responsible for teaching, advising students, monitoring academic performance, recording attendance, and providing academic support.
Interviewee	Lecturer Representative (Director of CFGS and Lecturer from SEHS)
Date of Interview	June 12, 2026
Method	Face-to-face semi-structured interview, audio recorded with signed consent.
Interview Recording	https://drive.google.com/file/d/1_mE1hMN7X9MDWJ7N-UxXoqegOp5ziEn2/view?usp=drive_link
Interview Transcript	https://drive.google.com/file/d/1cyCMno5Ugkk1WOGGWyvXtwe9ZUcTtF-Z/view?usp=drive_link
Interview Transcript (with time stamps)	https://drive.google.com/file/d/1UOf5Sg0mhuyRwgEnJOeP3Vm9TzyZfMXR/view?usp=drive_link
Consent Form and Interview	https://drive.google.com/drive/folders/1YzUVerH5dY7UyOQvAGHu9rsSMobYytw4?usp=drive_link

FR-L05	The system shall display Course Learning Outcome (CLO) achievement statistics for students.	Should Have	Interview	Proposed
FR-L06	The system shall automatically detect timetable conflicts during course registration.	Must Have	Interview	Proposed
FR-L07	The system shall notify students of timetable clashes before registration is completed.	Must Have	Interview	Proposed
FR-L08	The system shall recommend suitable courses based on academic history and previously failed courses.	Should Have	Interview	Proposed
FR-L09	The system shall notify students when prerequisite courses have not been completed.	Should Have	Interview	Proposed
FR-L10	The system shall allow lecturers to view advising records for students under their supervision.	Must Have	Interview	Proposed
FR-L11	The system shall display student attendance statistics and campus engagement indicators.	Should Have	Interview	Proposed
FR-L12	The system shall generate alerts for students who are academically at risk.	Must Have	Interview	Proposed
FR-L13	The system shall allow lecturers to add comments and academic recommendations to student reports.	Should Have	Interview	Proposed
FR-L14	The system shall maintain a complete academic history for each student from admission until graduation.	Must Have	Interview	Proposed
FR-L15	The system shall allow students to upload academic and administrative documents to their profile.	Should Have	Interview	Proposed

Part 3: Non-Functional Requirements

Appendix 3: HASNAIN HAIDER - Management Interview Resources

Resource / Evidence	Link or Location	Visual Evidence
Management/staff resource folder	Open resource folder	

AIU Student Hub

Stakeholder: AIU Management and Staff

Part 1: Stakeholder Information

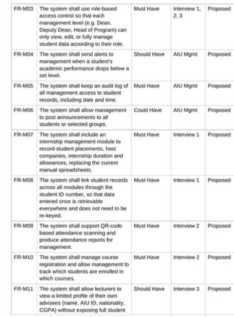
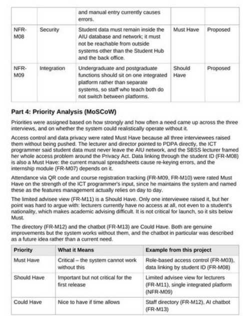
Field	Details
Stakeholder Group	AIU Management and Staff
Role in System	University administrators, academic staff and ICT staff who oversee and support academic and operational activities through the Student Hub.
Interviewee 1	Mr. Abubakar – Lecturer, School of Computing and Informatics (SCI); Director, Center for Community and Industry Engagement; SCI Internship Coordinator (approx. 2 years at AIU)
Interviewee 2	Mr. Muhammad Ruhaifi bin Zainol – ICT Programmer, AIU (approx. 3 years, involved in Student Hub development)
Interviewee 3	Dr. Siti Aznor binti Ahmad – Teaching Staff, School of Business and Social Sciences (SBSS) (joined February 2026)
Date of Interviews	June 2026
Method	Face-to-face, audio recorded with signed consent

Note: In line with the signed consent forms, interviewees should be referred to only by role (e.g. "AIU Management Representative") in the final group report. Names are kept here for internal traceability only.

Part 2: Functional Requirements

From the initial template emerged FR-M01 through FR-M06, later validated as pertinent in interview discussions. FR-M07 to FR-M13 were added based on what the three interviewees actually said.

Req. ID	Requirement Description	Priority (MoSCoW)	Source	Status
FR-M01	The system shall provide a dashboard for management to view student enrollment, attendance, and academic performance.	M	AIU Mgmt	Proposed
FR-M02	The system shall allow management to generate and download student performance reports by semester or faculty.	Must Have	AIU Mgmt	Proposed

Management functional requirement evidence	Included in management resource folder	
Management priority and ethical evidence	Included in management resource folder	

FR-M03	The system shall use role-based access control so that each management level (e.g. Dean, Deputy Dean, Head of Program) can only view, edit, or fully manage student data according to their role.	Must Have	Interview 1, 2, 3	Proposed
FR-M04	The system shall send alerts to management when a student's academic performance drops below a set level.	Should Have	AIU Mgmt	Proposed
FR-M05	The system shall keep an audit log of all management access to student records, including date and time.	Must Have	AIU Mgmt	Proposed
FR-M06	The system shall allow management to post announcements to all students or selected groups.	Could Have	AIU Mgmt	Proposed
FR-M07	The system shall include an internship management module to record student placements, host companies, internship duration and allowances, replacing the current manual spreadsheets.	Must Have	Interview 1	Proposed
FR-M08	The system shall link student records across all modules through the student ID number, so that data entered once is retrievable everywhere and does not need to be re-keyed.	Must Have	Interview 1	Proposed
FR-M09	The system shall support QR-code based attendance scanning and produce attendance reports for management.	Must Have	Interview 2	Proposed
FR-M10	The system shall manage course registration and allow management to track which students are enrolled in which courses.	Must Have	Interview 2	Proposed
FR-M11	The system shall allow lecturers to view a limited profile of their own advisees (name, AIU ID, nationality, CGPA) without exposing full student	Should Have	Interview 3	Proposed

		and manual entry currently causes errors.		
NFR-M08	Security	Student data must remain inside the AIU database and network; it must not be reachable from outside systems other than the Student Hub and the back office.	Must Have	Proposed
NFR-M09	Integration	Undergraduate and postgraduate functions should sit on one integrated platform rather than separate systems, so staff who teach both do not switch between platforms.	Should Have	Proposed

Part 4: Priority Analysis (MoSCoW)

Priorities were assigned based on how strongly and how often a need came up across the three interviews, and on whether the system could realistically operate without it.

Access control and data privacy were rated Must Have because all three interviewees raised them without being pushed. The lecturer and director pointed to PDPA directly, the ICT programmer said student data must never leave the AIU network, and the SBSS lecturer framed her whole access problem around the Privacy Act. Data linking through the student ID (FR-M08) is also a Must Have: the current manual spreadsheets cause re-keying errors, and the internship module (FR-M07) depends on it.

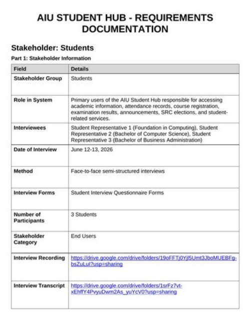
Attendance via QR code and course registration tracking (FR-M09, FR-M10) were rated Must Have on the strength of the ICT programmer's input, since he maintains the system and named these as the features management actually relies on day to day.

The limited advisee view (FR-M11) is a Should Have. Only one interviewee raised it, but her point was hard to argue with: lecturers currently have no access at all, not even to a student's nationality, which makes academic advising difficult. It is not critical for launch, so it sits below Must.

The directory (FR-M12) and the chatbot (FR-M13) are Could Have. Both are genuine improvements but the system works without them, and the chatbot in particular was described as a future idea rather than a current need.

Priority	What it Means	Example from this project
Must Have	Critical – the system cannot work without this	Role-based access control (FR-M03), data linking by student ID (FR-M08)
Should Have	Important but not critical for the first release	Limited advisee view for lecturers (FR-M11), single integrated platform (NFR-M09)
Could Have	Nice to have if time allows	Staff directory (FR-M12), AI chatbot (FR-M13)

Appendix 4: S M NADIM MAHMUD - Student Interview Resources

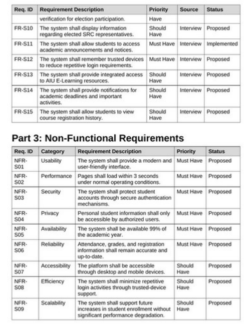
Resource / Evidence	Link or Location	Visual Evidence
Student interview resource folder	Open resource folder	
Interview recording	Open recording folder	

AIU STUDENT HUB - REQUIREMENTS DOCUMENTATION

Stakeholder: Students

Part 1: Stakeholder Information

Field	Details
Stakeholder Group	Students
Role in System	Primary users of the AIU Student Hub responsible for accessing academic information, attendance records, course registration, examination results, announcements, SRC elections, and student-related services.
Interviewees	Student Representative 1 (Foundation in Computing), Student Representative 2 (Bachelor of Computer Science), Student Representative 3 (Bachelor of Business Administration)
Date of Interview	June 12-13, 2026
Method	Face-to-face semi-structured interviews
Interview Forms	Student Interview Questionnaire Forms
Number of Participants	3 Students
Stakeholder Category	End Users
Interview Recording	https://drive.google.com/drive/folders/19oFFTj0Yjl5Umt3JboMUEBFg-bsZuLuI?usp=sharing
Interview Transcript	https://drive.google.com/drive/folders/1srFz7vt-xEhffY4PvyuDwm2As_yuYcV0?usp=sharing

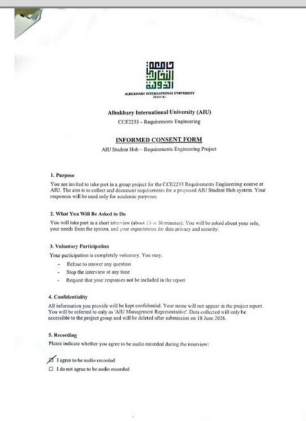
Interview transcript and timestamped transcript	Open transcript folder	
Consent form and interview form	Open consent/interview forms	

Req. ID	Requirement Description	Priority	Source	Status
	verification for election participation.	Have		
FR-S10	The system shall display information regarding elected SRC representatives.	Should Have	Interview	Proposed
FR-S11	The system shall allow students to access academic announcements and notices.	Must Have	Interview	Implemented
FR-S12	The system shall remember trusted devices to reduce repetitive login requirements.	Must Have	Interview	Proposed
FR-S13	The system shall provide integrated access to AIU E-Learning resources.	Should Have	Interview	Proposed
FR-S14	The system shall provide notifications for academic deadlines and important activities.	Should Have	Interview	Proposed
FR-S15	The system shall allow students to view course registration history.	Should Have	Interview	Proposed

Part 3: Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority	Status
NFR-S01	Usability	The system shall provide a modern and user-friendly interface.	Must Have	Proposed
NFR-S02	Performance	Pages shall load within 3 seconds under normal operating conditions.	Must Have	Proposed
NFR-S03	Security	The system shall protect student accounts through secure authentication mechanisms.	Must Have	Proposed
NFR-S04	Privacy	Personal student information shall only be accessible by authorized users.	Must Have	Proposed
NFR-S05	Availability	The system shall be available 99% of the academic year.	Must Have	Proposed
NFR-S06	Reliability	Attendance, grades, and registration information shall remain accurate and up-to-date.	Must Have	Proposed
NFR-S07	Accessibility	The platform shall be accessible through desktop and mobile devices.	Should Have	Proposed
NFR-S08	Efficiency	The system shall minimize repetitive login activities through trusted-device support.	Should Have	Proposed
NFR-S09	Scalability	The system shall support future increases in student enrollment without significant performance degradation.	Should Have	Proposed

Shared Group Resources

Resource / Evidence	Link or Location	Visual Evidence
Consent form template used by the group	Open consent form template	
GitHub repository highlighting everyone's efforts	[Placeholder - insert GitHub repository link manually]	

Albukhary International University (AIU)

CCE2233 – Requirements Engineering

INFORMED CONSENT FORM

AIU Student Hub – Requirements Engineering Project

1. Purpose

You are invited to take part in a group project for the CCE2233 Requirements Engineering course at AIU. The aim is to collect and document requirements for a proposed AIU Student Hub system. Your responses will be used only for academic purposes.

2. What You Will Be Asked to Do

You will take part in a short interview (about 15 to 30 minutes). You will be asked about your role, your needs from the system, and your expectations for data privacy and security.

3. Voluntary Participation

Your participation is completely voluntary. You may:

- Refuse to answer any question
- Stop the interview at any time
- Request that your responses not be included in the report

4. Confidentiality

All information you provide will be kept confidential. Your name will not appear in the project report. You will be referred to only as 'AIU Management Representative'. Data collected will only be accessible to the project group and will be deleted after submission on 18 June 2026.

5. Recording

Please indicate whether you agree to be audio recorded during the interview:

☑ I agree to be audio recorded

☐ I do not agree to be audio recorded